

Skill Arena - Phase 1 Part 3

PvP Arena, Match Flow, Replay & Disconnect Specification

Chapter 6 - PvP Arena Overview

PvP Arena is the core competitive system of Skill Arena. All future games must integrate with the PvP framework. Players compete using the same rules, same challenge seed and same starting conditions.

Match Creation

A match may be created through: - Ranked Queue - Casual Queue - Friend Challenge - Cross-League Challenge - Tournament Match Server generates Match ID and validates entry requirements.

Entry Validation

Server validates: - Account Status - Wallet Balance - League Eligibility - Ban Status - Tournament Qualification Client never performs final validation.

Match Pot Calculation

Example: Player A Entry = 10 Tokens Player B Entry = 10 Tokens Total Pot = 20 Tokens Platform Fee = 10% Platform Revenue = 2 Tokens Winner Receives = 18 Tokens

Match Start Process

Server: 1. Creates Match ID 2. Locks Entry Tokens 3. Generates Shared Challenge Seed 4. Stores Audit Record 5. Launches Match Both players receive identical conditions.

PvP Fairness Rules

Both players receive: - Same Maze - Same Difficulty - Same Lives - Same Timer - Same Objectives
No player may receive an advantage.

Live Match Information

Visible: - Your Progress - Your Timer - Your Lives - Opponent Progress - Opponent Lives Hidden: - Opponent Route - Opponent Exact Position - Opponent Solution Path

Opponent Visibility

Optional opponent preview: - Pixelated View - Delayed Feed - Progress Percentage Purpose: Create excitement without enabling copying.

Replay Engine

Every match recorded. Stored Data: - Match ID - Players - Start Time - End Time - Moves - Lives Lost - Progress - Winner - Payout Retention: Permanent.

Replay Access

Players may: - Review Historical Matches - Compare Routes - Share Replays Admins may: - Audit Results - Investigate Fraud - Resolve Disputes

Disconnect Handling

When a player disconnects: 1. Match Pauses 2. Both Screens Blur 3. Opponent Notified 4. Email Sent 5. 60 Second Timer Starts Failure to return = Forfeit.

Forfeit Rules

Forfeit results in: - Match Loss - Opponent Victory - Pot Distribution Repeat abuse may trigger penalties.

Challenge Contracts

Players may create custom challenges. Examples: - Higher Stakes - Cross-League Challenge - Friendly Match - Private Match Both parties must accept.

Spectator Framework

Future Feature: - Tournament Viewing - Championship Viewing - Replay Library Live viewing must never reveal solution paths.

Match Audit Trail

Every match creates: - Audit Record - Financial Record - Replay Record - Security Record Used for transparency and compliance.

Server Authority

The server controls: - Match Results - Pot Calculations - Replay Data - Win Conditions - Payouts
Client controls only: - Input - Display - Audio